

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination (New)

University Theory Examination March-April 2018

PH215 - Pharmacology-I

Date: 29/03/2018, Thursday

Time: 1.30 p.m. to 4.30 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. What does the term "bioavailability" mean?
[A] Plasma protein binding degree of substance
[B] Permeability through the brain-blood barrier
[C] Fraction of an uncharged drug reaching the systemic circulation following any route administration
[D] Amount of a substance in urine relative to the initial dose
 2. The characteristic structure of G-protein-coupled receptor is made up of
[A] Nine transmembrane alpha helices
[B] Seven transmembrane alpha helices
[C] Three transmembrane beta helices
[D] Five transmembrane beta helices
 3. What is characteristic of the intramuscular route of drug administration?
[A] Only water solutions can be injected
[B] Oily solutions can be injected
[C] Opportunity of hypertonic solution injections
[D] The action develops slower, than at oral administration
 4. Conjugation is:
[A] Process of drug reduction by special enzymes
[B] Process of drug oxidation by special oxidases
[C] Coupling of a drug with an endogenous substrate
[D] Solubilization in lipids
 5. What does "affinity" mean?
[A] A measure of how tightly a drug binds to plasma proteins
[B] A measure of how tightly a drug binds to a receptor
[C] A measure of inhibiting potency of a drug
[D] A measure of bioavailability of a drug
 6. Indicate the location of M2 cholinoreceptor type:
[A] Heart
[B] Glands
[C] Smooth muscle
[D] Endothelium

7. Indicate a muscarinic receptor-blocking drug:
[A] Scopolamine
[B] Pipecuronium
[C] Trimethaphan
[D] Pilocarpine
8. Which of the following sympathomimetics acts indirectly?
[A] Epinephrine
[B] Norepinephrine
[C] Ephedrine
[D] Methoxamine
9. Which of the following sympathomimetics is a beta2-selective agonist?
[A] Terbutaline
[B] Xylometazoline
[C] Isoproterenol
[D] Dobutamine
10. Which of the following drugs is a nonselective beta receptor antagonist?
[A] Metoprolol
[B] Atenolol
[C] Propranolol
[D] Acebutolol
11. One of the following drug is an example of Prodrug
[A] Gentamicin
[B] Levodopa
[C] tetracyclin
[D] Morphin
12. What is the toxic metabolite of cyclophosphamide?
[A] Acroliene
[B] Fluoride
[C] Phosphamide
[D] Tetracyclophosphamide
13. Which food product is avoided with use of tetracycline?
[A] Wheat flour
[B] Soyabean
[C] Milk
[D] Eggs
14. What is the effect of combining adrenaline with local anaesthetic injection?
[A] Absorption of anaesthetic increases
[B] Vasoconstriction slows absorption of anaesthetics and prolongs local effect
[C] Vasodilation & reduce local effect
[D] No effect of anaesthetic agent observed
15. What are the consequences of drug interaction between warfarin and aspirin?
[A] Toxicity to liver
[B] Heart attack
[C] Risk of bleeding
[D] Rise in blood pressure
16. Tick the second messenger of G-protein-coupled (metabotropic) receptor:
[A] Adenyl cyclase
[B] Sodium ions
[C] Phospholipase C
[D] cAMP

17. Which of the following is example of G protein coupled receptor (GPCR)?
 - [A] Muscarinic receptor
 - [B] Nicotinic receptor
 - [C] Cytokine receptor
 - [D] Oestrogen receptor
18. Indicate the adrenoreceptor antagonist drug, which is a rauwolfia alkaloid:
 - [A] Prazosin
 - [B] Propranolol
 - [C] Reserpine
 - [D] Phentolamine
19. One of the following is an example of anticholinesterase drug
 - [A] Physostigmine
 - [B] Loratidine
 - [C] Adrenaline
 - [D] Metoprolol
20. Which of the following is not a injectable route of drug administration?
 - [A] Subcutaneuos
 - [B] Intravenous
 - [C] Intrathecal
 - [D] Sublingual

SECTION – II

- Q.2** Attempt any TWO of the following
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|---|--|----|
| A | Describe in brief about Parasympatholytic drugs along with side effects. | 05 |
| B | Write in brief about Parasympathomimetics | 05 |
| C | Write a detail note on actions of Atropine. | 05 |
- Q.3** Attempt any TWO of the following
- | | | |
|---|---|----|
| A | Classify Sympathomimetic drugs with examples | 05 |
| B | Classify Sympatholytic drugs in detail with examples. | 05 |
| C | Compare and contrast Alpha and Beta adrenergic blocking agents. | 05 |

SECTION-III

- Q 4** Attempt any FOUR of the following
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|---|--|----|
| A | Describe Pharmacokinetic type of drug interaction | 05 |
| B | Explain in short about following terms: Pharmacokinetics and Pharmacodynamics. | 05 |
| C | Write a short note on Bioavailability | 05 |
| D | Provide a short note on Plasma protein binding and Half life | 05 |
| E | Differentiate: Ganglion blockers vs Neuromuscular blockers | 05 |
| F | Provide a brief note on local anesthetics | 05 |

Q 5 . Attempt any FOUR of the following

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|----------|--|-----------|
| A | Explain the mechanism of G-protein coupled receptor | 05 |
| B | Explain the concept of Agonist and Antagonist | 05 |
| C | Discuss preclinical testing in new drug development | 05 |
| D | Write a short note on Dose response relationship | 05 |
| E | Write a short note on drug metabolism | 05 |
| F | Explain in brief about drug excretion | 05 |
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